Development and evaluation of a metascience scraping tool

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Abstract

In the scientific community there is a growing concern of false results being published. Researchers are discussing if there is an ongoing replication crisis, with many claiming that there is. Several large replication studies have been performed with results demonstrating a large number of studies that cannot be replicated with significant results. This may be due to publication bias and p-hacking, where significant study findings are favored with publishers. Significance is found by excessive testing on a single data set. P-hacking needs to be quantified but the issue is more commonly discussed than studied. A reason is that it can be difficult to gather the large amounts of data necessary to demonstrate statistical irregularities. Manually collecting data on test statistics is both time consuming and complex. The problem addressed in this study is this difficulty and the need to integrate scattered data. A means of automating data collection is web scraping by collecting data from different websites converting it into a structured format. An already existing web scraping solution for a single scientific journal is expanded in this study. The research question posed is *Can the existing web scraping solution be expanded while maintaining the same precision?*. In order to answer the research question design science is utilized as a research strategy and an artifact is created. The focus is put on the design science phases development and evaluation. During development different aspects of the existing solution were reused, but it was rewritten using the Python framework Scrapy. Ex-ante evaluations were performed throughout the development process for iteratively testing and verifying the functionality of the feature solution. During the evaluation of the artifact the metric precision was used. An extensive ex-post evaluation was performed, manually annotating 4817 values from 200 articles. This data set was compared to that gathered by the artifact, resulting in a precision of 96.12 percent. This was a slight improvement to what the existing solution resulted in, which was 95 percent. It was concluded that the solution fulfilled the research goal, and while a single web scraping solution cannot be created to scrape all academic journals it was demonstrated how the existing solution could be expanded.

Keywords: Web Scraping, Scrapy, HTML, Metascience, P-values, P-hacking, Publication Bias

Synopsis

Background

There is a growing concern that a large share off published research is not replicable meaning that results cannot be duplicated when studies are repeated. Related to this is publication bias and p-hacking. A consequence of this is an increased distrust in science and the hindering of scientific progress. Quantifying p-hacking is important by analyzing published p-values. To do so a large amount of data needs to be collected. A means to automatically extract data is web scraping. It is an umbrella term for the collection of data from one or several websites, converting the data into a structured format.

Problem

Performing quantitative metascience studies requires large amounts of data. The difficulty of data gathering has hindered such studies. Manually gathering data is time-consuming and complex. Data is also often scattered. The problem addressed in this study is the difficulty of manual data collection in metascience studies.

Research Question

An existing solution hard coded for the Economic Journal is expanded in this study. The research goal is to scrape data from more journals while maintaining the precision of the existing solution. In this study it is investigated how statistical values can be collected, and the posed research question is: *Can the existing web scraping solution be expanded while maintaining the same precision?*

Method

The research strategy utilized in this study is design science. It is an appropriate choice of strategy as an artifact is created addressing a specific problem. It was also created aiming to support people in a practice by enabling quantitative metascience studies. The focus is put on two specific design science research activities, development and evaluation, as an outline of the problem and requirements already existed. When developing the artifact, the Python

framework Scrapy was used as it is applicable for large projects. When evaluating the artifact precision was used as a metric. Data was collected manually, to be compared to the result of the artifact.

Result

The existing solution was expanded with four more journals to a total of five journals, following the same statistical rules to extract different values. The Scrapy framework was used to create a spider for each journal. An extensive ex-post evaluation was performed. A total of 200 articles and 4817 p-values were evaluated. In this several possible improvements were identified, which were not discovered during development as the articles used in ex-ante evaluation did not exhibit those characteristics. The precision differed slightly across journals, but the total precision was 96.11 percent. The journal with the highest precision had the largest amount of published values and was the journal most extensively evaluated ex-ante.

Discussion

The artifact demonstrated how a web scraping solution can be expanded. As per the research question, precision was nearly maintained, but it was also shown that differences in publication standards slightly lowered it. A solution was created for different journals published in HTML. A limitation is that those available in PDF or JSON were excluded due to the inability to automatically identify table areas. This may affect how the collected data set can be generalized across the discipline. The focus was put on economic journals and when analyzing the data conclusions can only be drawn across the included study designs.

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List of Abbreviations

API - Application Programming Interface
BS4 - Beautiful Soup 4
CAPTCHA - Completely Automated Public Turing test to tell Computers and Humans Apart
CFAA - Computer Fraud and Abuse Act
CI - Confidence Interval
CSS - Cascading Style Sheet
CSV - Comma Separated Values
DoS - Denial of Service
DSR - Design Science Research
EDR - Expected Discovery Rate
ERR - Expected Replication Rate
FP - False Positive
HTML - Hypertext Markup Language
HTTP - Hyptertext Transfer Protocol
NHST - Null Hypothesis Significance Testing
ICA - Inter-Coder Agreement
JSON - JavaScript Object Notation
SE - Standard error
TP - True Positive
TPR - True Positive Rate
W3C - World Wide Web Consortium
Xpath - XML Path Language

Chapter 1

Introduction

An introduction is given to the field of metascience and the pending replication crisis spread over various disciplines in science. The connected concepts p-hacking and publication bias are presented to provide insight into ongoing concerns within metascience. Web scraping and its areas of application is introduced.

1.1 Background

1.1.1 Metascience and the replication crisis

Ioannidis (2005, p. 696) stated that "there is increasing concern that in modern research, false findings may be the majority or even the vast majority of published research claims". The author claims that it can be proven that a lot of published results are false and Szucs and Ioannidis (2017) stated that false reporting of probability in published papers is likely higher than 50 percent. While some researchers are more optimistic (Peterson & Panofsky 2020), Baker (2016) demonstrated in a survey that 52 percent of 1576 researchers questioned consider there to be a significant reproducibility crisis in science, and 36 percent a slight crisis. This corresponds to the fact that many scientific findings are not reproducible or replicable. The terms are sometimes incorrectly used synonymously. Sayre and Riegelman (2018) suggest that reproducibility refers to trying to duplicate the results of a study using the same materials and procedures. Replicability on the other hand involves using the same procedures to gather new data in order to try to duplicate the results. There have been several large reproducibility efforts made that have garnered a lot of attention. One is The Open Science Collaboration (2012) Reproducibility Project in psychology where 100 experiments were attempted to be reproduced, with only 36 percent of them being reproduced with significant results. Peterson and Panofsky (2020, p. 5) suggest several issues that contribute to the crisis; "increased competition, incentives that reward improbable findings, and lax policing of bad or fraudulent science".

With the attention of reproducibility and replicability in science, the metascience movement has seen an increase in related studies. Schooler (2014, p. 9) describes metascience as "an approach in which science turns the lens of scrutiny on itself". It is a transdisciplinary movement according to Peterson and Panofsky (2020, p. 3) seeking change science-wide. By producing quantitative studies that describe and evaluate science on a macro scale, reforms are motivated including "more elaborate reporting requirements, data deposition demands, and more replications". The authors discuss how the reproducibility crisis brought together different strands of metascience; open science activism, science of science, and statistical and methodological critiques. Open access was advocated by researchers before the crisis but has since gained more traction as possible errors may go uncorrected when no outside inspection is possible. Open science encourages researchers to publish their data, code, and details of analytical decisions. Science of science concerns quantitative studies of science, which has amplified as data availability has increased with the migration of research journals and data to online platforms. Furthermore, computational scholars migrate from different fields to perform science of science. Methodological and statistical critique can be traced back to before the reproducibility crisis, but it has reinvigorated, such as criticism of null hypothesis significance testing (NHST).

1.1.2 P-hacking and publication bias

According to Head et al. (2015), there is a growing concern within the scientific community that the focus on confirmatory and statistically significant results may lead to increased bias when publishing academic works. This is because significant study findings are more desirable for various reasons, such as career advancement. This is defined as publication bias. One form of such bias may be found regarding the null hypothesis.

The null hypothesis states no association between variables or that an experimental modification has no impact. Fisher (1925) developed NHST to objectively differentiate relevant data from background noise. Because the NHST method utilizes an arbitrary cutoff value, typically 0.05, it has led to the problematic habit of considering "significant" findings more meaningful, dependable, and repeatable. Many researchers have called for the abolition of NHST. In contrast, others have pointed out that many of the same issues with publication bias can be found in alternative methods, such as reporting effect sizes and confidence intervals.

Smith and Ebrahim (2002) state that the focus on NHST for "significant" findings has led to the advent of so-called p-hacking. The significance is artificially manipulated to reach a higher level of significance; this is performed through brute-force searching where multiple NHSTs are tested upon a single dataset until significance (above cutoff value) may be discovered.

According to Colquhoun (2017) one consequence of p-hacking is the increased distrust in science it may cause for the general public as well as the academic community. Head et al. (2015) states that publication of false positive results

may hinder scientific progress, and quantifying p-hacking is thus important. Colquhoun (2017) argues that universities and journals must cease undermining attempts to increase reproducibility by putting unfair incentives on those who perform the job.

1.1.3 Web scraping

A means to automatically extract data is web scraping. Web scraping has a long history, dating back almost to the inception of the internet. In 1993 the first automated web crawler was deployed by then MIT student Matthew Gray with the intention of measuring the size of the World wide web (*Bio for Matthew Gray* n.d.). Web scraping is an umbrella term for the automated collection or harvesting of data from one or several websites followed by conversion into basic formats like spreadsheets, databases, or CSV files for further processing or analysis. A wide variety of applications for web scraping exists. One is to collect research data that may be used to prove or disprove a research question. Others may include training material for machine learning models. The process may be performed either manually or automated. It is not only a complicated process, but it is also resource and time-intensive, especially when done manually (Diouf et al. 2019). Furthermore, there are several technologies that can be used for web scraping. Sirisuriya et al. (2015) describe Hypertext Transfer Protocol (HTTP) programming where data is retrieved by sending requests to a remote web server. Another technique is Hypertext Markup Language (HTML) parsing where HTML pages are parsed and content is retrieved and transformed.

1.1.4 Web data extraction areas of application

According to Mitchell (2018), there exist many applications for web scraping technologies, the primary one being the ability to collect and process vast amounts of data quickly. The most common way for people to access and gather information from the internet is through a web browser. Websites are built for being easily navigated with usability in mind, or at least most web designers emphasize this as their primary design philosophy. However, when collecting data from a large number of websites, the process soon becomes laborious for the user tasked with gathering the information. Hence, there are clear advantages to automating the process to save time and resources that may be used elsewhere. In most scenarios, if the content is available to be viewed in a web browser, it may be gathered with a Python script instead as data on websites follows an organized structure in the form of HTML. The use of automated scraping tools is almost as old as the internet itself, although it has been more commonly known as screen scraping or web harvesting. An example of web scraping being used in a real-world scenario is the 2006 project named "We feel fine". An automated scraping tool was developed for retrieving text from blogs where the blogger expressed their feelings. The data was collected and could then be visualized in real-time on a web page, giving the visitors a summary of how people around the world were feeling at the time. The example may be

seen as pointless, but data collected in the same manner is today used in more beneficial use cases such as medical research.

1.2 Problem

As pointed out by Peterson and Panofsky (2020), researchers performing quantitative metascience studies depend upon large amounts of data to demonstrate statistical irregularities and to draw insights from the scientific process. Zuckerman (2020) mention that misconduct in science, such as publication bias and p-hacking, is more often discussed than it is studied quantitatively. The author suggests that out of several reasons one might be the difficulty in launching new data gatherings. However, it is further pointed out that opportunities for quantitative metascience studies exist and should be pursued. There are several difficulties related to manual screening of research. Glez-Peña et al. (2014) highlight that it is time-consuming, demands expertise in the specific field and details are easily missed.

Web scraping can solve these issues. However, information is often scattered, and there is generally a need to consult several heterogeneous sources, which is a difficulty for web scraping. Nayak et al. (2021) discuss that heterogeneity is due to different formats such as HTML, PDF, and tables. Moreover, identifying precise data is further complex when the data needed to be extracted resides in different, or even unknown, places. Mitchell (2018) point out that if an application programming interfaces (API) for data extraction exists that is generally preferable to use to attain a stream of well-formatted data. However, such an API does not always exist when a set of data is to be collected over a substantial collection of websites, which is the case for certain journals.

Therefore, the problem addressed is the difficulties of manual data collection in metascience studies and the need to integrate data from heterogeneous and scattered sources.

1.3 Existing Solution

In this thesis, an already existing unpublished solution for scraping the Economic Journal will be expanded. It was originally developed by Sonja Remmer with a background in statistics and economics during the years 2020 to 2021. Remmer worked as a research assistant at the Stockholm School of Economics, where the solution was developed in collaboration with Professor Anna Dreber. Documentation of the previous solution's algorithm, the Python code, and data from the evaluation rounds have been shared for the continuation of the development of the algorithm.

1.4 Research Goal

The existing solution is hard coded for a specific journal, The Economic Journal, which publishes HTML formatting papers. However, the scraping tool is desired to be expanded to work for different document types and journals. Furthermore, the precision of the tool is to be maintained, meaning that the correct values are identified and collected. In the final evaluation round for the existing solution, precision was the most critical metric, more so than recall, resulting in 95 percent precision of collected values.

1.5 Research Question

This research will investigate how statistical values may be collected and analyzed from academic journals. The posed research question is:

Can the existing web scraping solution be expanded while maintaining the same precision?

Chapter 2

Extended background

This section aims to give an extended background of this thesis for the discussed concepts and research.

2.1 Statistical measures in metascience

Turner et al. (2019) discuss how the misuse of p-values has been criticized in the field of science. In metascience, there has been a call for changing guidelines related to p-values. Efron (2012, p. 32) describes p-values as "a universal language for hypothesis testing". While agreeing that the measurement has been overused and abused, the author argues that it has brought order to scientific reporting. Turner et al. (2019) mention that The American Statistician released a statement regarding p-value with several principles. It was said that scientific conclusions should not be based only on if a p-value is below a certain threshold, nor does it measure the effect or importance of a result. The measurement can possibly indicate if some data is incompatible with a specific statistical model but is in isolation not a good measure of evidence to a hypothesis.

Correspondently, Colquhoun (2017) agrees that p-values are commonly misinterpreted and states that p-values of 0.05 equal a rather high likelihood of false-positive results, higher than 5 percent as is suggested by the value. In several metascience studies, there is a recommendation that the threshold should be lowered if p-values are to be used. Vidgen and Yasseri (2016) suggests a threshold of 0.01 or 0.001, while Benjamin et al. (2018) suggest 0.005. Colquhoun (2017) criticizes the fact that 0.05 is labeled as statistically significant in several scientific disciplines and considers it to contribute to the reproducibility issue. In fact, both Colquhoun (2017) and Benjamin et al. (2018) do not think that definitions such as significant, very significant and not significant should be used at all in relation to p-values. It is true that the lower the p-value the less likely it is that the null hypothesis occurs, however, Colquhoun (2017) states that there is no easy way to say precisely how small a p-value should be. Despite this criticism the author believe that researchers should continue to use p-values and

confidence intervals, but to complement it with for example reverse Bayesian method to achieve a more secure false positive rate. Furthermore, this is how the measurement was intended to be used when first defined by Fisher (1925). Nuzzo (2014) describes that it was never meant to be used as a definitive test, but rather as a means of identifying data worthy of a second look.

Another statistical approach used in metascience is computing Z-scores and Z-curves. Bartoš and Schimmack (2020) state that it can be used to produce the expected replication rate (ERR) and the expected discovery rate (EDR) for a set of published studies. The measures are based on mean power and are used for assessing the credibility of published works. Bartoš and Schimmack (2020, p. 2) refer to EDR as "the percentage of statistically significant results that were obtained in all studies", or in other words estimate the rate of false positives. It can also provide an estimate of successful replication rates, as similar to ERR. Alexander and Green (2021) examined evidental value in a selection of articles by analysing the results with both p-curves and Z-curves. With both methods the authors found evidental value, but the Z-curve analysis showed a higher indication of publication bias. As p-curve analysis has received criticism, Huang et al. (2021) encourages the development of statistical tools to counteract this, which Z-curves is an example of. Furthermore, Brunner and Schimmack (2020) state that Z-curves perform better than p-curves when the data set it heterogeneous in terms of effect sizes. However, McAuliffe et al. (2021) point out that Z-curves are a relatively new method and interpretation of its results should be made with some caution.

2.2 Related research in metascience

Several large replication studies have obtained a lot of attention in both the scientific community, as well as in the public eye. Turner et al. (2019) argue that due to the issues identified by replication studies, it is warranted to perform more replications, and for it even to become an expected ethical practice. The authors state that there are scientific disciplines where it is not yet. confirmed if there are replication issues or not. Camerer et al. (2018) discuss that systematic evidence of reproducibility has been scarce up until recently. However, "The Reproducibility Project: Psychology" by Open Science Collaboration (2012) where 100 studies were reproduced ignited the scientific debate on the subject. Camerer et al. (2018) performed their own replication study where 21 high powered experimental studies were replicated with analysis plans reviewed by the original authors. It was found that significant results could be replicated in 62 percent of the studies. Furthermore, it was concluded that the effect size of all replications was on average around 50 percent, and 75 percent among the successful replications. The authors argue that this may suggest a systematic bias in estimated effect sizes in published articles.

Raff (2019) posed a question about what makes research independently reproducible within the field of machine learning, referring to trying to reproduce results without the original code. While open-source code and data are pro-

moted, it is suggested that it may be insufficient when reproducing results. If results cannot be reproduced even though the code is not disclosed, that may suggest issues with the paper such as "insufficient explanation of the approach, failure to describe important minute details, or a discrepancy between the code and description" (Raff 2019, p. 1). The author tried to reproduce 255 published papers independently succeeding in 63.5 percent of the cases. Furthermore, a number of features from the papers were selected and it was calculated how well they corresponded to a successful reproduction. The most significant features found were readability, algorithm difficulty, pseudo code, compute needed, number of equations.

Turner et al. (2019) discusses that one direction in metascience is to study the extent of issues such as p-hacking and publication bias by statistical analysis by extracting data from published work. Brodeur et al. (2016) collected 50,078 hypotheses tests from 641 different articles. The authors stated that the large number of data collected was necessary to achieve the conclusions of their study. The authors intended to study distribution of p-values. They were able to observe a two-humped pattern in the distribution which is believed to be an inconsistency related to researchers behavior; p-hacking, and the selection process of the journals; publication bias. Head et al. (2015) note that when p-value curves are studied and p-values are identified below 0.05 it is not possible to distinguish if it depends on p-hacking or selective publication bias. Brodeur et al. (2016) describes the two-humped pattern as that a first bump in the curve signifies where p-values are commonly represented, then a valley in the curve, and finally a larger bump just below the p-value threshold of 0.05. Unless p-values just below 0.05 are the most favored ones, this indicates that tests that produce significance slightly above the threshold may have been reanalyzed to produce a lower p-value. There is a very low representation of marginally insignificant results. The authors found that the pattern can be regularly found over different journals and years. It does appear to be more pronounced in journals using asterisk symbols as eye-catchers for marking significance.

Brodeur et al. (2020) found that p-hacking and publication bias appear to vary by method of inference. The authors studied more than 21,000 hypotheses from 25 different economic journals. They identified the same two-humped pattern as Brodeur et al. (2016). Furthermore Brodeur et al. (2020) found that p-hacking and publication bias did not appear to differ in different journals, including those considered to be the leading ones.

Head et al. (2015) used a text-mined data set to perform p-curve analysis also to study the distribution of p-values across different disciplines. The authors found evidence for p-hacking in p-values closer to 0.05. There was also a larger amount of published p-values in this range than lower values considered more consistent with evidential value. However, Bishop and Thompson (2016) stated that using p-curve analysis feasible to estimate the extent of p-hacking, or publications with evidential value, is not always appropriate. The authors studied the text-mined data set of Head et al. (2015) and features of the underlying data in a heterogeneous set of papers may not be represented in an automated data set. This means that p-curves might be erroneously produced.

Huang et al. (2021) discuss that p-curve analysis does not provide information about the general population but rather only the sample it is used on. However, the larger the number of studies it is used on the more generalizable it would be.

2.3 Web scraping process

According to Glez-Peña et al. (2014) the process of web scraping can be divided into three main approaches; site access, content extraction, and output building, the first being establishing access to the site from which the user intends to collect data. Access is established technically through communication with the target web site through the HTTP protocol and through compliance adhering to the terms of use often outlined in the `robots.txt` file hosted on the site's web server. These files are the implementation of The Robots Exclusion Protocol which is a standard used by websites to communicate information to web crawlers about areas of websites they are disallowed to crawl (*The Web Robots Pages* 2007). The terms of use dictate in what manner data may be gathered from the site by automated means, such as the number of requests being sent to the web server in order to avoid overloading or overuse of system resources.

Glez-Peña et al. (2014) states that when access has been established, HTML parsing and contents extraction may be performed. This process aims to extract data of interest with techniques such as regular expressions and additional algorithmic logic. HTML based websites follow the document object model (DOM) structure, a programming interface for web documents. Witch simplifies the parsing of data as values and data reside within a defined format. The final step of the process is data building, where the now extracted data can be made into a structure and can be analysed further. Often the data is stored in text-based formats such as XML or CSV files.

2.4 Web scraping techniques

Germann (2018) describe that there are several generic web scraping applications in existence that does not require any programming. However, they lack functionality in scraping targeted data. In contrast, Python as a programming language excels at this task with a long list of useful libraries and frameworks. Two of the most common techniques are Beautiful Soup and Scrapy.

2.4.1 Beautiful Soup 4

Beautiful Soup 4 (BS4) is a Python HTML parsing library for navigating, searching, modifying and extracting data from web formatted documents. This is done by converting documents to Unicode and then parsing it into navigable trees (Germann 2018). Zhao (2017) discuss that BS4 is a convenient technique for scraping. When parsing HTML, it can automatically detect the encoding of the parsing and subsequently convert it to a client-readable encode. However,

Hajba (2018) point out that BS4 is not an HTTP client, and a library such as Requests must be used to extract data from URLs. The author describes BS4 as easy to use, nonetheless, in contrast to the framework Scrapy, BS4 can be difficult to optimize to avoid bottlenecks. Furthermore, it ignores web sites `robots.txt`.

2.4.2 Scrapy

Scrapy is an open-source web crawling and scraping Python framework. Where Beautiful Soup is a library that parses HTML, Scrapy makes it possible to write spiders that crawl websites from a base URL and subsequently extract structured data from them by using selectors (*Scrapy 2.6.1 documentation* 2022). The framework is known for being robust and efficient. While having a steeper learning curve than BeautifulSoup, Germann (2018) argue that it is appropriate to use in large and ongoing projects. Zhao (2017) discuss that using Scrapy speeds up the process in such cases and adds that it is usable for scalable and maintainable scrapers. Shi et al. (2016) concur and further add that advantages include Scrapy understanding broken HTML code, and there being continuous stability fixes. Hajba (2018) goes as far as to say that Scrapy is the only viable Python tool able to handle complex scraping tasks as it offers a range of possibilities in terms of configuration.

The Scrapy process is depicted in 2.1. Shi et al. (2016) discuss that there are five major components; the Scrapy Engine, the Spider, the scheduler, the item pipeline, and the downloader. The Scrapy Engine is the core and in control of the data flow. The Engine receives URLs to scrape from the Spider, and schedules them as requests in the Scheduler. The requests are then passed on to the Downloader via the Engine through potential Downloader middleware. The Downloader communicates with the Internet through the HTTP protocol. The Downloader generates responses containing HTML source code after downloading documents which it sends back to the Engine. The Spider handles the responses, and information is then stored in items.

XPath

XPath is a built-in mechanism in Scrapy. Based on the tree representation of XML or HTML documents, it allows navigation around this representation (Dale 2016). It is built on the library lxml (Alshammari et al. 2020). XPath has different nodes that in turn has different relationships to each other, such as parent, child and sibling. Path expressions are used to select the desired nodes (*XPath Introduction* n.d.). An expression is evaluated to yield an object with one of four types; a node-set, a boolean, a floating-point number, or a string (Clark et al. 1999).

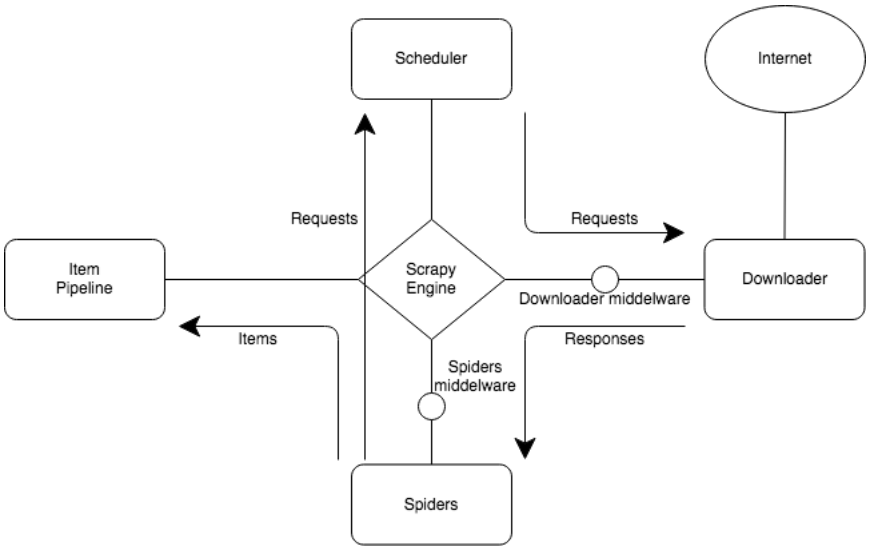

Figure 2.1: Scrapy Process

CSS-Selectors

Alshammari et al. (2020) mention that another navigation option is CSS (Cascading Style Sheet) Selectors. Both Xpath and CSS-Selectors are W3C (World Wide Web Consortium) standards. As with XPath, it is built on lxml and is similar in terms of performance and accuracy. HTML is targeted by name, tags, classes, and attributes. As described in *Selectors* (n.d.), pattern matching determines which CSS rules apply to which elements in the document tree. The patterns are referred to as selectors. If all conditions of the selector is true for an element it is considered to be a match.

2.5 Web data extraction challenges

Krotov and Silva (2018) discusses legal aspects regarding web scraping. The legal theories and rules that govern web scraping include copyright infringement, where the material being collected may fall under copyright legislation and thus is not permitted to be scraped. Another possible legal theory is breach of contract, where the scraping process violates the terms of service of the source being scraped. Services often want to limit the use of scraping tools as they may use a lot of system resources (such as bandwidth, web server workload). The Computer Fraud and Abuse Act (CFAA) may also apply to web scraping as the computer being accessed may be considered to be protected.

Furthermore, Krotov and Silva (2018) also discusses ethical aspects of web scraping. Web scraping can injure what the authors refer to as "sentient creatures" affiliated with a website, such as a website's owners or consumers, in

addition to breaking current laws. By definition, these adverse outcomes are difficult to anticipate. One example of an adverse effect could be the Diminishing Value for the Organization. As the web scraping application enters the website without utilizing the human-friendly web interface, they will not be exposed to the adverts used to commercialize its content but still use system resources in terms of bandwidth and server capacity, causing a loss in value for the website operator.

Meschenmoser et al. (2016) discuss measures that website operators employ to stop or limit the use of web scrapers accessing their content and taking up unwanted system resources. One of these measures is access barriers. The user is challenged with a CAPTCHA (Completely Automated Public Turing test to tell Computers and Humans Apart), a filter intended to be solvable by humans but difficult to be artificially solved by an algorithm. This is made to tell computers and humans apart, limited access for scraping software but maintaining the ability for legitimate users to access the content. Luscombe et al. (2021) further discuss possible defense mechanisms that might be encountered while web scraping. It can be disallowment defined in a website's `robots.txt`, banning IP addresses that send too many requests, user-agent blocking, and requirement of an API key.

2.6 Web data extraction in metascience research

Cosentino et al. (2016) discuss how web scraping technologies may be applied in a metascience context in order to collect data from various computer science conferences. The paper outlines the creation of a conceptual schema, where data in the form of a variety of conference-related information is collected, such as bibliographic metadata, conference websites, digital libraries, and quality indicator services. The collected data is then stored in a queryable database, where further analytics and insights may be gathered. The information is being collected incrementally, meaning that the update mechanism for the system is triggered at any moment when new data is made available, keeping the database up to date. The system's goal is to better understand and give further insights into conference research activities, thus an extensive collection of metrics is produced. From these metrics, useful information may be elicited, such as gauging the actual interest of topics by the community this is useful for future conferences, mainly for deciding upon what areas to cover next during feature conferences. Furthermore, more advanced analysis was also performed, revealing more in-depth insights, the team were able to reveal discrepancies between conference topics and the actual trend in the paper topics. Which may point to a misalignment between what is published in research papers regarding the exact topic of research, and it serves as a great example of how web scraping and analytic tools may be applied within metascience research.

Smith-Unna and Murray-Rust (2014) describe that technical and political barriers have inhibited mining academic publications at a scale. Several publishers have APIs that restrict the possibility to data-mine even though they make

their content available online. However, the authors state that web technologies make it possible to mine the publications regardless of the source. A web scraping tool stack was developed that loops through URLs and scrapes article metadata. Martone et al. (2016) demonstrated how the tool can be used in practice in combination with a second tool used to annotate academic publications. In combination automation of extraction of facts were performed which resulted in making literature review processes cheaper, more rigorous, continuous and transparent.

Jager and Leek (2014) published an analysis of p-values which they collected using text mining. However, the resulting data set received criticism from Ioannidis (2014) for exhibiting several weaknesses. Jager and Leek (2014) themselves pointed out that the fact that they only collected p-values from abstracts was a limitation. Ioannidis (2014) echoed this stating that p-values in abstracts are extremely selected and not representative of all p-values. Furthermore, it was not recognized that false-positive rates vary across disciplines and study designs. The data set produced predominantly included study designs with the lowest false-positive rates. The sample of both journals and papers were also highly selected. These factors make the data set non-generalizable. From the result of their analysis Jager and Leek (2014) stated that the rate of false discoveries 14 percent, a far lower number than other researchers have claimed. Ioannidis (2014) refuted this claim by stating that combining the wrong methods and unreliable data put a rest to their claims. Head et al. (2015) attempted to counter the weaknesses that were pointed out and text mine p-values from different disciplines. The authors found evidence of p-hacking across all disciplines.

Chapter 3

Methodology

The research strategy and method utilized is presented in this chapter. Design science was deemed most suitable for the project to produce an artifact possible to solve the stated problem and answer the research question.

3.1 Choice of strategy

3.1.1 Design Science

Johannesson and Perjons (2014) describe design science research (DSR) as a research method aimed at solving practical problems of general interest. If that is not possible, it should at least bring the researcher(s) closer to solving the stated problem. Meaning it focuses on creating an artifact containing knowledge and facilitating measuring the performance of said artifact that is being developed. An artifact may be a physical object but may also be models, methods, constructs, and instantiations that function in a more extensive system.

The design science process consists of a five-step methodology that is performed in an iterative manner. First, the problem is explicated, meaning it concretize what it intends to solve to the reader. Then requirements for the solution are elicited, which are functions the artifact must fulfill and exhibit when completed. In the design and development phase, suitable design ideas for the artifact are chosen, and a design prototype is constructed. A demonstration may then be conducted where stakeholders of the project may witness the artifact performing its function(s). Finally, one or more evaluations are conducted. This may also be performed during any of the phases in the design science process as its iterative. This is performed on the artifact to verify its function, feedback is given, and improvements may then be made (Johannesson & Perjons 2014).

Hevner et al. (2004) discusses DSR in information systems research and argues that the paradigm will play an essential role within system research and development. From a management perspective, the authors suggest that information system managers stand to benefit the most as they are actively engaged

with design activities and the creation and deployment of IT systems that enable the organization to meet the goals and requirements set. There are several examples of studies utilizing design science to develop artifacts related to web scraping. Liu et al. (2021) developed a framework for idea recommendation where web scraping was an integral part. Corresponding with design science the authors exploit theories and knowledge in the research area upon which the artifact was developed. Mobini and Karunaratne (2019) followed DSR methodology when eliciting requirements and selecting modules in order to construct an artifact consisting of dashboards displaying non-cognitive skills in academic curriculums.

Johannesson and Perjons (2014, p. 7) describe artifacts "as something that should support people in practice". The tool developed in this paper aims to do so by giving further insights in metascience research, and DSR is an appropriate methodology for achieving this. In this project, an already existing artifact will be extended in functionality. In DSR, this can be characterized as an improvement according to Johannesson and Perjons (2014). The problem area is known and a new solution is proposed. According to Hevner et al. (2004, p. 82) the most important guideline in DSR is to produce an "artifact created to address a problem". As the artifact developed in this project is done so with a specific problem in mind DSR is a usable choice of strategy.

3.1.2 Alternative research strategy

An alternative research strategy to design science is action research. However, action research was dismissed as it is less suitable for producing artifacts, and it places more focus on collaboration with practitioners than necessary in this thesis. Denscombe (2014) describes it as having a practical orientation aimed at solving existing problems. Similar to design science, action research is an iterative process involving a feedback loop for continuous improvements of what is being implemented. Iivari and Venable (2009) discuss that action research is context dependent with a dual goal of contributing to research and practice. Denscombe (2014) reiterates the link between research and practice. Instead of being treated as two different processes, they are integrated, and practitioners are highly involved throughout the research project. While participation promotes harvesting practitioners' knowledge, it can also limit the scope and scale of the research and limit generalizations. Practitioners are consulted in the development of the artifact in this project, but not to the degree intended in action research. In this sense the methodology of design science is more suitable.

Johannesson and Perjons (2014) point out that action research does not always include an artifact as part of the solution to a practical problem. Instead, it can involve psychological, social, and organizational change. Furthermore, Iivari and Venable (2009) exemplifies this by stating that in a case where the research interest lies in building and evaluating a technological artifact, design science is more applicable than action research. The desired outcome of this project was a practical tool, much in line with what can be produced as output from a design science research project.

Should action research have been used as the primary research strategy for the thesis, some adjustments would have been made to the methodology. Firstly, a more tight-knit corporation with the stakeholder would be required with weekly meetings and progress reports to create a tighter feedback loop between the stakeholders where small iterative adjustments could be made. Furthermore, this approach would enable closer integration with the artifact and the organizational environment it aims to be used within.

3.2 Choice of method

Commonly, DSR projects emphasize one or two of the five activities defined by Johannesson and Perjons (2014). A typical case in DSR is development- and evaluation-focused research. The development starts from an existing problem and requirements specification. In this project, further development of an already existing web scraping tool was made. The problem and requirements were outlined in advance by the original developer. The requirements have been refined throughout development of the tool, but it has not been the main focus of the project.

3.2.1 Develop artifact

The core of DSR is to design and develop the artifact. Peffers et al. (2007) describe that the development of an artifact using DSR should use existing theories and knowledge to produce a solution. Johannesson and Perjons (2014) state that the knowledge base upon which the artifact is built may include artifacts similar to the one being created. For the artifact developed knowledge from both the field of metascience and computer science were sought and used as an influencer. Peffers et al. (2007) outline that during this DSR phase the desired functionality and architecture is expressed, and subsequently the actual artifact is developed. As described by Johannesson and Perjons (2014) the input to this activity is an outline of the desired artifact and a set of requirements.

3.2.2 Evaluate artifact

Hevner et al. (2004) state that artifacts should be evaluated in respect to the usefulness provided in terms of solving the previously defined problem. According to Peffers et al. (2007) the objectives of the artifact are compared to observed results.

It is also an opportunity to identify improvements in further design. This was done as a formative evaluation. It was iteratively evaluated during the development of the artifact to identify possible improvements continuously. Ex-ante evaluations were carried out to do so, as these are appropriate when the artifact is not yet fully developed. Ex-ante refers to evaluating the artifact before it is fully developed. This was done by testing certain aspects of the tool on a limited number of articles. It is emphasized by Hevner et al. (2004) that

design is inherently an iterative process. Thus, the development phase is dependent on the evaluation phase where feedback is provided in cycles. However, ex-ante evaluations cannot produce reliable results as summative evaluations (Johannesson & Perjons 2014).

Apart from ex-ante evaluations, there was also a need to assess the quality of the final product to conclude how useful the artifact is. Therefore, a summative evaluation was chosen to be performed after completing the development phase. Johannesson and Perjons (2014) describe this as the assessment of the final version of the artifact. The focus of this study is to crawl and collect values from academic papers, and to evaluate to what extent the tool collects the correct values. Therefore, precision was used as the evaluation metric for the summative ex-post evaluation performed.

3.3 Application of method

The application of the method can be categorized into two main activities, the development of the artifact and the evaluation of the artifact. The time expenditure for completing these two main activities is estimated to be a 70/30 split, where the development of the artifact composes the former and the evaluation of the results produced in the latter.

Requirements were elicited during discussions with professor Anna Dreber. Details regarding which journals and data to scrape was decided. The existing solution was also analyzed to conclude what data to scrape.

3.3.1 Develop Artifact

Technical Aspects of The Existing Solution

The existing solution developed by Sonja Remmer has been developed in the Python programming language. The language has well-established support for third-party libraries. This is utilized to a great extent, which is not uncommon for web scraping solutions in general. Currently, the libraries used are; Requests, Pandas, BS4, NumPy, lxml and SciPy. Requests facilitate HTTP connectivity for Python, making it more user friendly. Pandas is used for data manipulation and analysis of the gathered data. BS4 is the main library used for the scraping process. It enables the collection of entire web pages in HTML format, which may be parsed with lxml. Parsing in this context is the creation of a nested data structure from the collected HTML data. SciPy is used to compute the calculations for the solution, such as P and Z-Values used in conjunction with SciPy is NumPy, a library used to enable and operate on large multidimensional arrays.

The solution also makes heavy use of regular expressions, which is a programming technique used for specifying search patterns for text and searching through text and, for this scenario, the scraped HTML content—identifying tables that are of interest for processing and collection of values, as well as identifying content that is not of interest and may be disregarded.

Applied Scraping Framework

As the goal for this thesis is to create a solution that works for several journals it was chosen to use the framework Python Scrapy. It offers advantages as being scalable, maintainable, and applicable for larger projects. Furthermore, Scrapy offers a range of possibilities to configure the spider. It was necessary to throttle speed in order to scrape articles in an ethical and responsible manner, and to not risk the spider being banned. Scrapy was therefore deemed as more suitable for this project, rather than continue development using Beautiful Soup.

3.3.2 Evaluate Artifact

Evaluation Metrics

Precision is a performance metrics used in the evaluation of effectiveness in information retrieval. The precision metric is used to determine the proportion of positive identifications that was actually correct (*Precision and recall machine learning crash course, Google Developers* 2020). This was done by scraping approximately 400 articles from four journals. Out of this data set 200 articles were randomly selected and P- and Z-values were manually annotated and compared to the scraped data set.

The recall metric is used to identify the proportion of actual positives that were identified correctly (*Precision and recall machine learning crash course, Google Developers* 2020). The rules for collecting values used in developing the artifact focus on finding values belonging to the main hypothesis of the academic papers. Therefore, it would be valuable to evaluate recall in relation to how many values belong to the main hypothesis were collected. However, due to the scope and resources of this study this was not evaluated and the focus was instead placed on precision in line with the research question.

3.4 Ethical Considerations

Johannesson and Perjons (2014) discusses ethical guidelines and principles for conducting research, with the first principle being not to cause suffering or harm to any individual whether part of the study or not. This study will be based on analyzing published research from academic journals and does not intend to damage the reputation or cause any harm to the authors or other individuals.

Luscombe et al. (2021) discuss web scraping's value in relation to research and point out that it is coupled with ethical considerations for scientists. The authors mention that as of yet there is no ethical research policy regarding web scraping. Therefore, it is important that researchers themselves are aware of what can cause harm while sending requests, such as overwhelming web servers and unintentionally causing denial of service (DoS) attacks. Kouzis-Loukas (2016) argue that throttling of speed should therefore be implemented. The author further discuss that irresponsible web scraping should not be conducted.

When web scraping a user-agent should be included to identify oneself. Furthermore, preferences such as request frequency expressed in `robots.txt` files should be respected and not overridden.

The settings file for the crawler in the project is set to obey the `robots.txt`. Furthermore, the settings were configured to auto-throttle the scraping speed taking into consideration factors such as download time, server response, and load. Auto-throttle is an extension in the Scrapy framework that dynamically adjusts the crawling speed in order to not overwhelm web servers. This can be regarded as ethical spidering.

Chapter 4

Results

4.1 Development

In table 4.1 the journals considered to be included in the study are listed. The Economic Journal is the journal that was scraped with the existing solution. Journals being published by for example Elsevier does not allow web scraping in their `robots.txt` but instead offers their own API. The JSON files that can be extracted are not structured to allow identification of table titles, tables and table footnotes which is necessary to extract the correct values. It was therefore excluded from the study. However, the journals listed with HTML format as being scrapable have explicitly allowed web scraping and are the ones the tool have been developed for.

Publisher	**Journal**	**Format**	**Scrapable**	**Scraped**
Oxford Academic	The Quarterly Journal of Economics	HTML	Yes	Yes
	The Economic Journal	HTML	Yes	Yes
The University of Chicago Press	Journal of Political Economy	HTML	Yes	Yes
Springer Science	Journal of Economic Growth	HTML	Yes	Yes
	Review of World Economics	HTML	Yes	Yes
Elsevier	Journal of Financial Economics	JSON	Yes	No

Table 4.1: List of journals

When developing the solution a set of articles were used from each journal to iteratively test the tool. These articles were not used in the ex-post evaluation. The Scrapy framework was used to create a spider for each journal. When running the spiders, it is necessary to be connected to an institutional network in order to have access to published articles. In the spiders URLs to the publications are defined as a starting point. A request is made to these URLs and a response object is passed to a default callback method called parse. The response object contains various parameters. For example, the HTML status code which defaults to 200 if nothing went wrong. Different nodes of the HTML-code can then be accessed using `response.xpath`. In the parse method

links to articles are looped through using XPath. The links are followed and all articles are looped through. For Journal of Economic Growth and the Review of World Economics links had to be followed to another depth as all tables in the articles had their own pages. XPath was used to select and extract different elements from the different articles. In order to for example determine which methods are used in the articles, whether the tables should be scraped or not, and which tables in the tables should be scraped helper methods and regular expressions are used. The rules regarding which statistical data to scrape was reused from Remmer's solution. Changes have been made to regular expressions and to how the method is collected due to different publication standards. The same statistical rules are used for all journals, but the XPaths are specific for each journal which is why different spiders were created.

In listing 4.1 examples of XPaths are given. The first line of code collects the DOI code from the requests response. The XPath filters by the descendant node "//meta" searching within the "@name" attribute for the string "dc.identifier", which contains the DOI link code that is to be extracted for the article. The second line collects the year from the article text and not the entire requests response, also by filtering by the descendant nodes "//a" and "//time", using "/text()" to get the content — continuing to search within the "@name" attribute for the href link "#article-info" link. The third and fourth lines extract the authors and the subject(s) in the same manner as the first and second lines. An important distinction is the ".getall()" ending attribute, which tells Scrapy to collect all of the elements' contents as there may be multiple values. The table footnotes are collected similarly but within the tables instead of the whole article. Collecting the table footnote is essential for deciding whether it should be scraped.

```
doi = response.xpath('//meta[contains(@name, "dc.identifier")]/@content').get()
year = article.xpath('//a[contains(@href, "#article-info")]//time/text()').get()
authors = article.xpath('//meta[contains(@name, "dc.creator")]').getall()
subject = article.xpath('//li[contains(@class, "c-article-subject-list__subject")]//span/text()').getall()
table_footnote = table.xpath('//div[contains(@class, "c-article-table-footer")]//li').get()
```

Listing 4.1: XPath example

In listing 4.2 an example of regular expressions used is given. After the table footnotes have been collected, regular expressions are used to determine whether the table is suitable for scraping. Where `"\b"` is used as a meta escape for asserting a word boundary position. The pipe sign is a logical or as multiple versions of the word (small starting letter etcetera) are interesting. The `"[^.]*"` quantifier matches the first group with the below group so that if within the same sentence, any version of the first and second group should trigger the if statement as true.

```
def identify_element_type_footnote(table_footnote):
```

```
    element_type = []
    if re.search(r'(\bSE\b|standard errors?|Standard errors?|Standard Errors?)[^.]*(parentheses|Parentheses|parenthesis|Parenthesis)', str(table_footnote)) or re.search(r'(parentheses|Parentheses|parenthesis|Parenthesis)[^.]*(\bSE\b|standard errors?|Standard errors?|Standard Errors?)', str(table_footnote)):
        element_type.append("standard_error")

    if re.search(r'(t[^.]?\s?statistics?|t[^.]?\s?values?)[^.]*(parentheses|parenthesis)', str(table_footnote), re.IGNORECASE) or re.search(r'(parentheses|parenthesis)[^.]*(t[^.]?\s?statistics?|t[^.]?\s?values?)', str(table_footnote), re.IGNORECASE):
        element_type.append('t_statistic')

    if re.search(r'(p[^.]?\s?values?)[^.]*(parentheses|parenthesis)', str(table_footnote), re.IGNORECASE) or re.search(r'(parentheses|parenthesis)[^.]*(p[^.]?\s?values?)', str(table_footnote), re.IGNORECASE):
        element_type.append('p_value')

    if re.search(r'(z[^.]?\s?statistics?|z[^.]?\s?values?)[^.]*(parentheses|parenthesis)', str(table_footnote), re.IGNORECASE) or re.search(r'(parentheses|parenthesis)[^.]*(z[^.]?\s?statistics?|z[^.]?\s?values?)', str(table_footnote), re.IGNORECASE):
        element_type.append('z_value')
```

Listing 4.2: Regular Expression example

In order to determine if values in tables should be scraped the table title and footnote are used. Tables with certain keywords are disregarded, and so were appendix tables. The reason is that values belonging to the main hypothesis are the ones that are aimed to be scraped. Figure 4.1 depicts a simplified process of how the tool decides if a table should be scraped or not, and which values it should scrape based on which type of statistical values exist in the table. In order, it checks if the footnote mentions either standard errors, t-statistics, p-values or z-statistics. All of these should be mentioned in the same sentence as parentheses. If none of the elements were found, it is finally checked if standard errors are mentioned without parentheses in the same sentence. If an element is found, the table is looped through in order to identify the rows with the values of interest and these are then scraped.

It is also checked if a table contains several different panels. If it does, this is seen as being several sub-tables published on top of each other and the first row from each panel is scraped. P-values are not always written out directly but are more commonly presented in regression tables with coefficients and standard errors. Z-statistics are then calculated by dividing coefficients by standard errors, followed by calculating p-values from z-statistics. Therefore, standard errors are looked for first, thereafter t-statistics, p-values and z-statistics.

If no values are collected from a table being scraped due to it not being possible to identify which values are published in the table based on the text in the

footnote, the tool instead discerns if the table potentially still contains interesting values. This is done by looping through the table and checking whether two asterisks (**) exist, or if standard errors are written in the table. If this is true, the table is marked in the output as potentially containing interesting values, but it is not scraped. An example of this is illustrated in appendix figure A.1.

After scraping an article, the following data is written to an Excel file with each row representing a value from a table:

(A) Journal

(B) DOI

(C) Year

(D) Number of authors

(E) Subject (JEL codes)

(F) Assumed method

(G) Potential alternative methods

(H) Article containing placebo tests

(I) Article containing potentially relevant values (if p/z were not scraped)

(J) Table row containing intercept

(K) Table number

(L) Placebo tests in table

(M) Robustness tests in table

(N) Z-statistics

(O) P-value

4.2 Evaluation

This section discusses the ex-post evaluation process of the data collected and extracted from the Journal of Political Economy. The data was collected manually from the table(s) deemed relevant, and the subsequent Z-values were calculated, and P-values were then derived from said values.

A total of approximately 100 articles were collected from each journal, out of these, 50 were randomly selected and manually annotated, divided into 25 per coder, with an overlap of a total of 10 pieces being used for validating the collection method and the inter-coder agreement (ICA) between the coders. In total, 200 articles have been evaluated.

Out of these 10 articles, 233 values were collected by each coder. Artstein and Poesio (2008) argue that if manually annotated data is to be used is development and testing there it is necessary to show the reliability of such data. Further, the authors mention that if the manual coders produce consistently similar results it demonstrates a unified understanding of annotation guidelines. It can therefore be expected that they will continue to perform in a consistent manner when annotating separate data. Ensuring this reliability of the annotated data set is a means for demonstrating validity of the evaluation. In this study it shows that the annotated data set is capturing true P- and Z-values.

According to Artstein and Poesio (2008), a simple measure of agreement between coders is to calculate the percentage of the agreement. The authors note that when using for example two or four categories to annotate the data set this measurement is not appropriate as it does not account for chance. However, in the case of this study the annotation is not set to categories since p-values are derived or manually calculated. Therefore, percentage of agreement is used in this case. Out of the 233 inter coded values the coders agreed on all of them. The agreement is therefore A = (233 + 233) / 466 = 1.0 (100 percent agreement). 222 true positives and 11 false positives were found in the inter-coded data set. This demonstrates reliability of the annotation guidelines and validity of the entire data set. However, it should be noted that there is a possibility that while the coders agree on the annotations they may both commit the same error. An excerpt from the output from the artifact can be seen in figure A.1. The annotated data set is stored and can be used as a future reference set.

An agreement of 100 percent is rare and it should be noted that the articles used for this metric only represent one of the journals that the tool have been developed for. This was the journal resulting in the highest precision, which could possibly indicate that articles from this journal are more straight forward. Thus, being easier to annotate and performing the best in the evaluation. While the 10 articles were randomly selected, there is of course a risk that they are not representative for the entire data set scraped by the tool. So, while the ICA score does add reliability to the evaluation, it should not be assumed that the score would be the same if all four journals were represented.

DOI	**Table**	**Scrape**	**Method**	**Z-value**	**P-value**	**Comment**
10.1086/701425	4	Yes	IV	1.998407643	0.0456723437	
10.1086/701425	4		IV	2.76	0.005780249	
10.1086/701425	4		IV	1.916413374	0.0553123634	
10.1086/701425	4		IV	2.576923077	0.0099684724	
10.1086/701425	4		IV	2.342175066	0.0191716801	
10.1086/701425	4		IV	4.173913043	0.0000299593	
10.1086/701425	5	No	IV			No SE
10.1086/701425	6	Yes	IV	2.400651466	0.0163659022	
10.1086/701425	6		IV	2.911564626	0.0035963697	

Table 4.2: Example of evaluation structure

Table 4.2 illustrates an example of the structure of the data collected. First, the digital object identifier for the article; second, the table number in descending order and whether it should be scraped or not, followed by the method for the study and subsequent P- And Z-values calculated. Lastly, optional comments were left by the annotator. For example, no values were collected from table 5 because no standard errors (SE) existed in the table.

After the data set was manually collected, it was compared to the data collected by the algorithm. If the values collected by the algorithm matched the value in the annotated data set, it was marked as a true positive. If it did not match it was marked as a false positive. The true positive rate (TPR) is calculated by dividing the number of true positives by the number of true positives and false positives (TP/TP+FP). In table 4.3 the number of true positives and false positives per journal are noted. Furthermore, the precision is calculated for each journal. The bottom row shows the total number of all values including the total precision score for all journals. While the same number of articles were annotated from each journal it can be seen that the amount of values published in each journal differs significantly. The journal with the highest precision of 97.9 percent, Journal of Political Economy, has the largest amount of values which may have affected the precision. Furthermore, it was the journal that was first used in developing the solution and most extensively evaluated ex-ante which is likely a reason for it exhibiting high precision. The worst precision was for the Journal of Economic Growth, 94.6 percent. When aggregating the results from all journals 4817 values were evaluated in total. Out of these, 187 values were marked as false positives, resulting in a total precision of 96.12 percent.

Journal	**TP**	**FP**	**TP+FP**	**Precision**
JPE	1640	35	1675	0.979
JEG	1413	80	1493	0.946
RWE	659	34	693	0.95
QJE	918	38	956	0.96
	4630	**187**	**4817**	**0.9612**

Table 4.3: True and false positive rates

A precision score was calculated for each article's manual annotation. From this the mean and standard deviation for each journal was derived. These are presented in table 4.4 along with confidence intervals (CI). In concurrence with Vidgen and Yasseri (2016), a significance value of 0.01 was used to calculate confidence intervals. For the Journal of Political Economy with the highest precision the mean was 0.97 with a standard deviation of 0.15. That resulted in a 99 percent confidence interval of 0.97±0.06. In the final row of the table an aggregation of the results is presented. With a mean of 0.93 it resulted in 99 percent confidence interval of 0.93±0.05.

An ex-ante evaluation was performed in order to improve the artifact iteratively, during the development of the artifact. Meaning a predefined set of

	Mean	**Stdev**	**99 % CI**
JPE	0.97	0.15	0.06
JEG	0.92	0.27	0.11
RWE	0.92	0.27	0.11
QJE	0.91	0.21	0.10
All	0.93	0.23	0.05

Table 4.4: Means, standard deviations and confidence intervals

articles belonging to a specific issue of a journal was selected. These articles were then evaluated iteratively against the output the artifact produced, and improvements and corrections were made accordingly to the issues and bugs discovered. Several possible improvements were then later identified during the more extensive ex-post evaluation. These improvements were not identified earlier as the same issues were not applicable to the articles used during the ex-ante evaluation. While recall was not used as an evaluation metric certain aspects of the artifact that affect recall were noted. The following needs for improvements were identified:

1. Corresponding coefficients and standard errors are usually located in the same column. When they are separated in different columns the algorithm has mistaken standard errors for coefficients and the wrong Z-value has been calculated resulting in a false positive.

2. When p-values and standard errors are written in brackets instead of parentheses the algorithm does not scrape them.

3. In some articles it was found that both standard errors and standard deviations were written in parentheses on the same row. The algorithm treated all as standard errors and calculated several false positives.

4. In certain tables where p-values are published in parentheses the algorithm has mistaken them for standard errors and calculated Z-values from them resulting in false positives.

5. Some minor bugs are present in the current version of the tool. While they are not affecting performance, they will also need to be corrected in future versions of the tool. Minor bugs include the spreading of tables in the output file where tables from the same article are spread out and do not uniformly print under the same row. Tables are also printed in descending order, which is not coherent with how tables are published in the articles scraped, where tables are printed in ascending order.

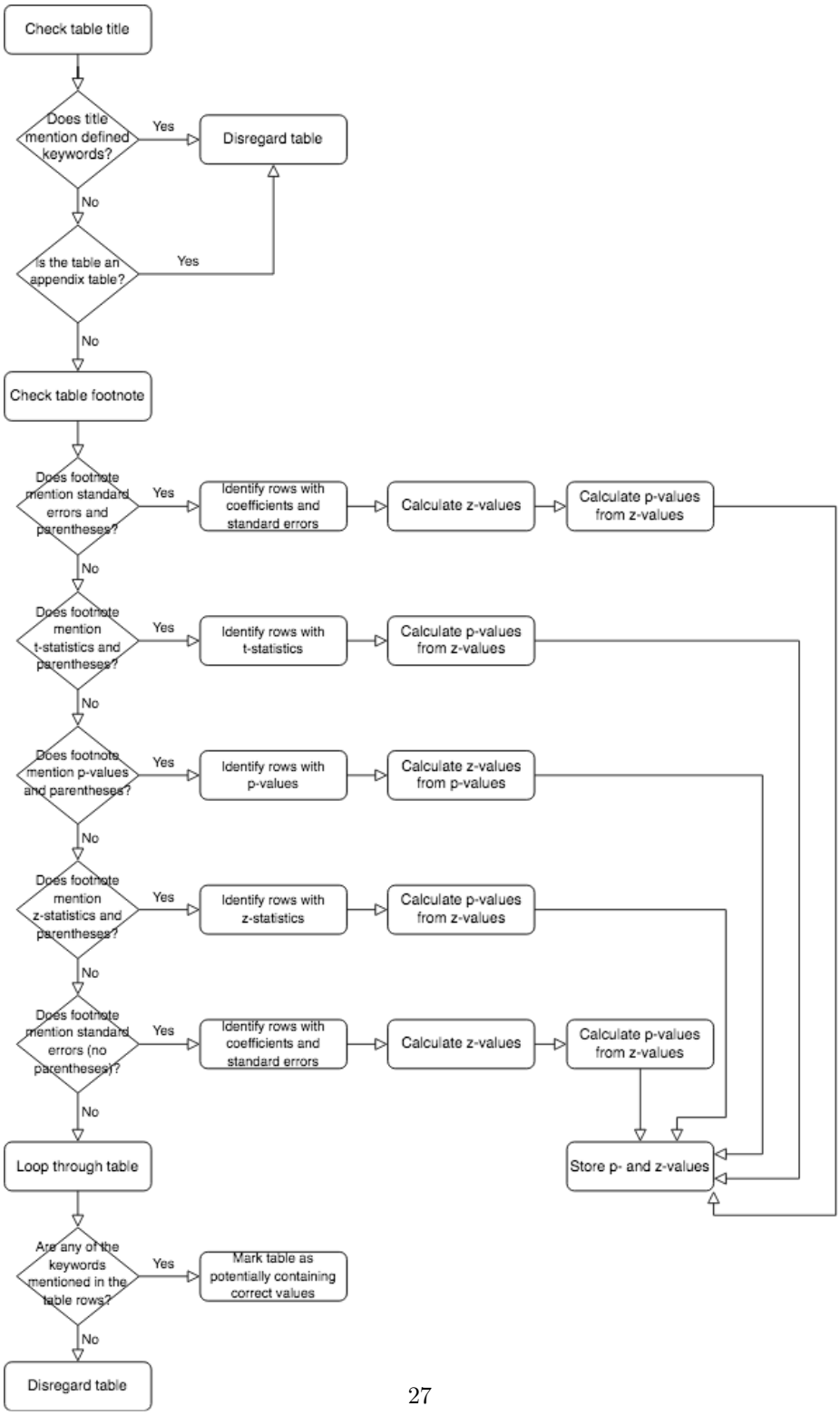

Figure 4.1: Extraction of statistical data flowchart

Chapter 5

Discussion and conclusion

5.1 Discussion

As both Diouf et al. (2019) and Glez-Peña et al. (2014) discussed, manually collecting data is both time-consuming and complex. The ex-post evaluation in this study was extensive including 200 articles and 4817 values. In terms of time, it took a lot of resources. As mentioned, probability scores are not always published as clear p-values which adds to the complexity of gathering them. In light of the evaluation, performing what Peterson and Panofsky (2020) refers to as science of science studies would be demanding without any automatic means of collecting data.

An early issue during development was that not all publishers allow web scrapers to crawl their different journals. As discussed by Krotov and Silva (2018) ethical aspects is one of the challenges that will be encountered when web scraping. Smith-Unna and Murray-Rust (2014) do mention that even though that some publishers introduce barriers to inhibit web scraping, it may still be possible to create a solution to work around the barriers. However, as per researcher ethics, the barriers were respected and no effort to scrape these journals was made. Some publishers, for example Elsevier, offer their own API to collect data from academic papers for research projects. The data could then be accessed in JSON format. However, tabular data was necessary to identify and scrape. The full articles in JSON were not structured in a way that made it possible to find and isolate the parts belonging to the tables. In contrast, some aspects of the data were identifiable as unique objects with keys such as metadata, publishing date, etc. Tables, unfortunately, were not. Making it very difficult for the scraping tool to identify these tables as they lack the document object model structure found in HTML/XML, where data tables are contained within specific element nodes. A similar issue was encountered when investigating if it would be possible to scrape tables from articles published in PDF. There are several libraries in Python available to parse PDF files. They are useful for extracting text, but less useful when there is a necessity to retain the

structure of tables. However, tables with statistical data in academic journals do not follow a simple structure but often have several panels, uneven columns and areas. There is no library capable to correctly extract any type of PDF tables without first defining for example the area. The decision was made to develop the solution for journals published in HTML with a publisher allowing web scraping, thus ignoring PDF and JSON formats. The motivation was that the development would be more effective resulting in more papers being possible to scrape and more p-values being compiled.

Solutions for four journals from three different publishers were created. A different scraper was created for each journal. Between different publishers the HTML-code differed, as expected, whereby a single solution could not be created. However, it was noted that journals with the same publishers had less differences in structure. Therefore, less time was required to adapt the solution between these journals. Publication standards differ between both journals and publishers which was also necessary to adapt the solution after. For instance, The Quarterly Journal of Economics and Journal of Political Economy use roman numerals for different sections, which was necessary not to mistake for the method instrumental variables (IV). Furthermore, the regular expressions used was adapted to each publisher as different statistical data was written in different ways. It should be noted that all journals scraped by the tool are economic journals. Publication standards and used methods likely differ more for journals in different fields. Ioannidis (2014) does note that false-positive rates differ across disciplines and study designs and that these should not by jointly analyzed. Therefore, enough data needs to be collected for different study designs and as the different methods are compiled with the tool, starting with journals in the economic field is reasonable.

As defined in the research goal and question the solution developed is an expansion of a web scraper working for a single journal. The focus has been on being able to collect a larger amount of statistical values. Another goal could have been to attempt to increase the precision of the tool by defining more rules by which the tool collects the values. However, this would have required more extensive evaluation rounds during development and there would have been less time to include more journals. Zuckerman (2020) pointed out that there are opportunities for quantitative metascience studies, but collecting enough data is a barrier. The artifact in this study demonstrates the possibility to automatically gather enough data to statistically prove whether p-hacking is an issue. As seen in the study by Brodeur et al. (2016) who collected 50,078 hypothesis tests from 641 articles there is a need to collect a large amount of data to produce valid results. This amount of data can only be collected if values from different journals are collected.

5.2 Conclusion

The stated research question was:

"Can the existing web scraping solution be expanded while maintaining the same precision?"

An already existing solution able to scrape one journal was further extended with four more journals. To answer the stated research question, an artifact was produced with a validated precision of 96.12 percent. Meaning, that the solution should be seen as having fulfilled its research goal — the discrepancy of 3.88 percent can be attributed to the improvements identified in section 4.2 for increasing the overall precision. The solution developed by Sonja Remmer produced a precision of 95 percent. That means that a precision increase of 1.12 percent was seen in the expanded solution.

Confidence intervals were calculated for each journal. For Journal of Political Economy, it can be concluded that with 99 percent confidence the precision of the algorithm is 0.97±0.06. However, the precision and confidence intervals of the algorithm varies between journals. For example, with the same confidence, The Journal of Economic Growth which had the worst score in the evaluation resulted in 0.92±0.11. This could indicate that the algorithm is biased towards different journals. There are several possible reasons for this. The main reason is likely the amount of time spent to the develop the tool for the different journals. Journal of Political Economy was the first journal that the tool was extended to, and most time was spent on it. The confidence interval for the tool and for an arbitrary journal of the four represented it is concluded that with 99 percent confidence the precision is 0.93±0.05. However, as presented, for every specific journal the confidence intervals are different.

Furthermore, some conclusions from the result can be made. There will never be a "do it all" generic solution for the scraping of academic journals with current scraping technologies. This type of data collection needs to be individually tailored for each journal. This, in turn, means that journals from the same publisher are faster to implement as there are more similarities between them.

5.3 Ethical and Social Consequences

As the artifact is yet to be deployed and used in a research scenario, it is hard to discern potential implications. However, as the evaluation has shown, the precision metric shows very close to the same levels as the original solution meaning that the tool can be expected to perform very similarly, or better, but is now available for other academic journals. Meaning the tool may be used in future quantitative studies. This is positive from a societal point of view as publication bias and p-hacking may be better understood and corrected. The implications of publication bias are discussed in this thesis, and the importance

of thwarting it should not be understated.

From an ethics standpoint, this thesis does not discuss the direct use of the tool but rather its development and technical aspects. No immediate ethical concerns for individual researchers can be directly argued for. For the sake of transparency, the data set collected during the evaluation phase of this thesis will be stored and made available upon request. However, the source code for the project may not be shared as it is to remain under closed source upon the request of the original developer.

5.4 Limitations

A limitation of the study is the exclusion of journals available in PDF or JSON formats, thus excluding statistical data from these in the data set possible to collect with the tool. This may affect how well conclusions drawn from analysis on the data set can be generalized across the discipline. Furthermore, as discussed the focus was put on economic journals. Analysis of the data collected with the developed tool reflect the study designs represented in the data set. It is not possible to generalize the data as to draw conclusions about different disciplines and study designs.

During the evaluation it was annotated if the method was correctly scraped, and if the p- and z-values were true positives or true negatives. Although the rules for whether values should be scraped or not are aimed at scraping those belonging to the main hypothesis, it was not evaluated whether the scraped values belonged to it due to the scope of the study and the need to evaluate data scraped from several journals. Bishop and Thompson (2016) argue that there is a necessity to know if the p-values relate to the main hypothesis of interest. This metric was evaluated by Sonja Remmer with a precision of 85 percent of true positive p-values belonging to the main hypothesis. Reliability and validity in terms of precision of the collected data has been addressed. However, in order to further prove the reliability of the collected data and to validate the truth of the result of the extended solution a more extensive evaluation including recall in relation to the main hypothesis would be necessary.

5.5 Future work

The developed tool in this thesis has provided the possibility to collect more data than the previous solution. As discussed, the HTML-code and underlying structure for journals with the same publisher is similar and to extend the solution with these would be a good start to further develop the tool. Eventually, more disciplines apart from economics can be included, but as discussed these should be kept separate during analysis.

Another approach to improve the tool would be to improve the precision of the collected values and to improve the precision of which values belong to the main hypothesis, including calculating the precision and recall as an

evaluation metric for gathering further insights and increasing precision. During the evaluation points relating to improving recall were identified even though it was not used as a metric. Improving precision and recall would require the definition of more rules of which tables to scrape by studying the evaluation performed.

A further extension of the tool that would require a different approach would be to find a solution to include publications in PDF and JSON formats.

Bibliography

Alexander, J. & Green, J. (2021), *Publication Bias and Evidential Value in Speech, Language, and Hearing Research*, OSF Preprints.

Alshammari, R., Alrashed, R., Almutiri, A., Alwalah, M., Al-marrai, W., Alqahtani, D. & Alzahrani, A. (2020), Data extraction based on web scrapy, *in* 'International Conference on Innovations in Bio-Inspired Computing and Applications', Springer, pp. 506–514.

Artstein, R. & Poesio, M. (2008), 'Inter-coder agreement for computational linguistics', *Computational linguistics* **34**(4), 555–596.

Baker, M. (2016), 'Reproducibility crisis', *Nature* **533**(26), 353–66.

Bartoš, F. & Schimmack, U. (2020), *Z-curve. 2.0: Estimating replication rates and discovery rates*, PsyArXiv.

Benjamin, D. J., Berger, J. O., Johannesson, M., Nosek, B. A., Wagenmakers, E.-J., Berk, R., Bollen, K. A., Brembs, B., Brown, L., Camerer, C. et al. (2018), 'Redefine statistical significance', *Nature human behaviour* **2**(1), 6–10.

Bio for Matthew Gray (n.d.). URL: https://www.mit.edu/ mkgray/bio.html, visited 2022-02-09.

Bishop, D. V. & Thompson, P. A. (2016), 'Problems in using p-curve analysis and text-mining to detect rate of p-hacking and evidential value', *PeerJ* **4**, e1715.

Brodeur, A., Cook, N. & Heyes, A. (2020), 'Methods matter: P-hacking and publication bias in causal analysis in economics', *American Economic Review* **110**(11), 3634–60.

Brodeur, A., Lé, M., Sangnier, M. & Zylberberg, Y. (2016), 'Star wars: The empirics strike back', *American Economic Journal: Applied Economics* **8**(1), 1–32.

Brunner, J. & Schimmack, U. (2020), 'Estimating population mean power under conditions of heterogeneity and selection for significance', *Meta-Psychology* **4**.

Camerer, C. F., Dreber, A., Holzmeister, F., Ho, T.-H., Huber, J., Johannesson, M., Kirchler, M., Nave, G., Nosek, B. A., Pfeiffer, T. et al. (2018), 'Evaluating the replicability of social science experiments in nature and science between 2010 and 2015', *Nature Human Behaviour* **2**(9), 637–644.

Clark, J., DeRose, S. et al. (1999), 'Xml path language (xpath)'.

Collaboration, O. S. (2012), 'An open, large-scale, collaborative effort to estimate the reproducibility of psychological science', *Perspectives on Psychological Science* **7**(6), 657–660.

Colquhoun, D. (2017), 'The reproducibility of research and the misinterpretation of p-values', *Royal society open science* **4**(12), 171085.

Cosentino, V., Cánovas Izquierdo, J. L. & Cabot, J. (2016), Metascience: An holistic approach for research modeling, *in* 'International Conference on Conceptual Modeling', Springer, pp. 365–380.

Dale, K. (2016), *Data visualization with python and javascript: scrape, clean, explore & transform your data*, " O'Reilly Media, Inc.".

Denscombe, M. (2014), *The Good research guide. Maidenhead, England*, McGraw-Hill/Open University Press.

Diouf, R., Sarr, E. N., Sall, O., Birregah, B., Bousso, M. & Mbaye, S. N. (2019), Web scraping: state-of-the-art and areas of application, *in* '2019 IEEE International Conference on Big Data (Big Data)', IEEE, pp. 6040–6042.

Efron, B. (2012), *Large-scale inference: empirical Bayes methods for estimation, testing, and prediction*, Vol. 1, Cambridge University Press.

Fisher, R. (1925), *Statistical Methods for Research Workers Oliver and Boyd, London. Reprinted in Statistical Methods, Experimental Design and Scientific Inference*, OUP, Oxford.

Germann, J. E. (2018), 'Approaching the largest 'api': extracting information from the internet with python', *Code4Lib Journal* (39).

Glez-Peña, D., Lourenço, A., López-Fernández, H., Reboiro-Jato, M. & Fdez-Riverola, F. (2014), 'Web scraping technologies in an api world', *Briefings in bioinformatics* **15**(5), 788–797.

Hajba, G. L. (2018), Using scrapy, *in* 'Website Scraping with Python', Springer, pp. 97–167.

Head, M. L., Holman, L., Lanfear, R., Kahn, A. T. & Jennions, M. D. (2015), 'The extent and consequences of p-hacking in science', *PLoS biology* **13**(3), e1002106.

Hevner, A. R., March, S. T., Park, J. & Ram, S. (2004), 'Design science in information systems research', *MIS quarterly* pp. 75–105.

Huang, Q., Peng, W. & Simmons, J. V. (2021), 'Assessing the evidence of perspective taking on stereotyping and negative evaluations: a p-curve analysis', *Group Processes & Intergroup Relations* **24**(8), 1306–1334.

Iivari, J. & Venable, J. R. (2009), 'Action research and design science research-seemingly similar but decisively dissimilar'.

Ioannidis, J. P. (2005), 'Why most published research findings are false', *PLoS medicine* **2**(8), 0696–0701.

Ioannidis, J. P. (2014), 'Discussion: why "an estimate of the science-wise false discovery rate and application to the top medical literature" is false', *Biostatistics* **15**(1), 28–36.

Jager, L. R. & Leek, J. T. (2014), 'An estimate of the science-wise false discovery rate and application to the top medical literature', *Biostatistics* **15**(1), 1–12.

Johannesson, P. & Perjons, E. (2014), *An introduction to design science*, Vol. 10, Springer.

Kouzis-Loukas, D. (2016), *Learning scrapy*, Packt Publishing Ltd.

Krotov, V. & Silva, L. (2018), *Legality and ethics of web scraping*, Emergent Research Forum (ERF).

Liu, Q., Du, Q., Hong, Y. & Fan, W. (2021), 'Idea recommendation in open innovation platforms: A design science approach', *China Center for Internet Economy Research (CCIE) Research Paper* .

Luscombe, A., Dick, K. & Walby, K. (2021), 'Algorithmic thinking in the public interest: navigating technical, legal, and ethical hurdles to web scraping in the social sciences', *Quality & Quantity* pp. 1–22.

Martone, M., Murray-Rust, P., Molloy, J., Arrow, T., MacGillivray, M., Kittel, C., Kasberger, S., Steel, G., Oppenheim, C., Ranganathan, A. et al. (2016), 'Contentmine/hypothes. is proposal', *Research Ideas and Outcomes* **2**, e8424.

McAuliffe, W. H., Edson, T. C., Louderback, E. R., LaRaja, A. & LaPlante, D. A. (2021), 'Responsible product design to mitigate excessive gambling: A scoping review and z-curve analysis of replicability', *PloS one* **16**(4), e0249926.

Meschenmoser, P., Meuschke, N., Hotz, M. & Gipp, B. (2016), 'Scraping scientific web repositories: Challenges and solutions for automated content extraction', *D-Lib Magazine* **22**(9/10).

Mitchell, R. (2018), *Web scraping with Python: Collecting more data from the modern web*, " O'Reilly Media, Inc.".

Mobini, P. & Karunaratne, T. (2019), Towards an ict enabler for enhancing non-cognitive skills in a lifelong learning setting, *in* 'European Conference on E-Learning; Kidmore End. Academic Conferences International Limited, Kidmore End, United Kingdom, Kidmore End', pp. 396–405.

Nayak, R., Balasubramaniam, T., Kutty, S., Banduthilaka, S. & Peterson, E. (2021), A semi-automatic data extraction system for heterogeneous data sources: a case study from cotton industry, *in* 'Australasian Conference on Data Mining', Springer, pp. 209–222.

Nuzzo, R. (2014), 'Statistical errors', *Nature* **506**(7487), 150.

Peffers, K., Tuunanen, T., Rothenberger, M. A. & Chatterjee, S. (2007), 'A design science research methodology for information systems research', *Journal of management information systems* **24**(3), 45–77.

Peterson, D. & Panofsky, A. (2020), 'Metascience as a scientific social movement'.

Precision and recall machine learning crash course, Google Developers (2020). URL: https://developers.google.com/machine-learning/crash-course/classification/precision-and-recall, visited 2022-04-01.

Raff, E. (2019), 'A step toward quantifying independently reproducible machine learning research', *Advances in Neural Information Processing Systems* **32**, 5485–5495.

Sayre, F. & Riegelman, A. (2018), 'The reproducibility crisis and academic libraries', *College & Research Libraries* **79**(1), 2.

Schooler, J. W. (2014), 'Metascience could rescue the 'replication crisis'', *Nature News* **515**(7525), 9.

Scrapy 2.6.1 documentation (2022). URL: https://docs.scrapy.org/en/latest/faq.html, visited 2022-03-26.

Selectors (n.d.). URL: https://www.w3.org/TR/CSS21/selector.htmlpattern-matching, visited 2022-03-29.

Shi, Z., Shi, M. & Lin, W. (2016), The implementation of crawling news page based on incremental web crawler, *in* '2016 4th Intl Conf on Applied Computing and Information Technology/3rd Intl Conf on Computational Science/Intelligence and Applied Informatics/1st Intl Conf on Big Data, Cloud Computing, Data Science & Engineering (ACIT-CSII-BCD)', IEEE, pp. 348–351.

Sirisuriya, D. S. et al. (2015), 'A comparative study on web scraping'.

Smith, G. D. & Ebrahim, S. (2002), 'Data dredging, bias, or confounding: They can all get you into the bmj and the friday papers', *Bmj* **325**(7378), 1437–1438.

Smith-Unna, R. & Murray-Rust, P. (2014), 'The contentmine scraping stack: literature-scale content mining with community-maintained collections of declarative scrapers', *D-Lib Magazine* **20**(11/12).

Szucs, D. & Ioannidis, J. P. (2017), 'Empirical assessment of published effect sizes and power in the recent cognitive neuroscience and psychology literature', *PLoS biology* **15**(3), e2000797.

The Web Robots Pages (2007). URL: https://www.robotstxt.org/robotstxt.html, visited 2022-03-28.

Turner, J. R., Brown, H. Q., Passmore, D. L., Nimon, K., Baker, R., Jeong, S. & Flatt, C. (2019), 'Metascience: Guidelines for the practitioner', *Advances in Developing Human Resources* **21**(4), 503–512.

Vidgen, B. & Yasseri, T. (2016), 'P-values: misunderstood and misused', *Frontiers in Physics* **4**, 6.

XPath Introduction (n.d.). URL: https://www.w3schools.com/xml/xpath$_{i}ntro.asp, visited 2022-03-29$.

Zhao, B. (2017), 'Web scraping', *Encyclopedia of big data* pp. 1–3.

Zuckerman, H. (2020), 'Is "the time ripe" for quantitative research on misconduct in science?', *Quantitative Science Studies* **1**(3), 945–958.

Appendix A

Output from artifact

An excerpt from the output when scraping the Review of World Economics is shown on the next page in this appendix. It is illustrated how for example different JEL codes and methods are scraped. It is also shown how some tables are marked to potentially contain interesting values (*). When a table is scraped it can be seen how both p- and z-values are written to the Excel document.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	journal	article	year	n authors	subject	method	ential_meth	acebo_artic	otential_tab	ential_inter	table	acebo_tabl	ustness_ta	z	p
2	rwe	doi:10.1007/s10290-021-00434-1	2021	3	O16, E44, F	macro		False	Table 4			False	False	*	*
3	rwe	doi:10.1007/s10290-021-00435-0	2021	2	C14, C26, E	IV		False	Table 4			False	False	*	*
4	rwe	doi:10.1007/s10290-021-00432-3	2021	2	F16, F22	other		False			Table 8	False	False	0,717718	0,472931
5	rwe	doi:10.1007/s10290-021-00432-3	2021	2	F16, F22	other		False			Table 8	False	False	0,931023	0,351842
6	rwe	doi:10.1007/s10290-021-00432-3	2021	2	F16, F22	other		False			Table 8	False	False	0,949505	0,342364
7	rwe	doi:10.1007/s10290-021-00432-3	2021	2	F16, F22	other		False			Table 8	False	False	0,951656	0,341272
8	rwe	doi:10.1007/s10290-021-00433-2	2021	2	F41, F43	other		False			Table 6	False	False	20,22622	5,75E-91
9	rwe	doi:10.1007/s10290-021-00433-2	2021	2	F41, F43	other		False			Table 6	False	False	32,07368	1E-225
10	rwe	doi:10.1007/s10290-021-00433-2	2021	2	F41, F43	other		False			Table 6	False	False	21,67402	3,6E-104
11	rwe	doi:10.1007/s10290-021-00433-2	2021	2	F41, F43	other		False			Table 6	False	False	10,35462	3,99E-25
12	rwe	doi:10.1007/s10290-021-00433-2	2021	2	F41, F43	other		False			Table 6	False	False	17,19753	2,77E-66
13	rwe	doi:10.1007/s10290-021-00431-4	2021	2	F10, F14, D	other		False			Table 6	False	False	6,25	4,1E-10
14	rwe	doi:10.1007/s10290-021-00431-4	2021	2	F10, F14, D	other		False			Table 6	False	False	8,666667	4,45E-18
15	rwe	doi:10.1007/s10290-021-00431-4	2021	2	F10, F14, D	other		False			Table 6	False	False	7,666667	1,77E-14
16	rwe	doi:10.1007/s10290-021-00431-4	2021	2	F10, F14, D	other		False			Table 6	False	False	8,5	1,9E-17
17	rwe	doi:10.1007/s10290-021-00423-4	2021	2	F10, F14, F	IV		False			Table 4	False	False	12,36364	4,11E-35
18	rwe	doi:10.1007/s10290-021-00423-4	2021	2	F10, F14, F	IV		False			Table 4	False	False	7,571429	3,69E-14
19	rwe	doi:10.1007/s10290-021-00423-4	2021	2	F10, F14, F	IV		False			Table 4	False	False	0,43	0,667196
20	rwe	doi:10.1007/s10290-021-00423-4	2021	2	F10, F14, F	IV		False			Table 4	False	False	11,4	4,18E-30
21	rwe	doi:10.1007/s10290-021-00423-4	2021	2	F10, F14, F	IV		False			Table 4	False	False	6,428571	1,29E-10
22	rwe	doi:10.1007/s10290-021-00423-4	2021	2	F10, F14, F	IV		False			Table 4	False	False	3,858824	0,000114
23	rwe	doi:10.1007/s10290-021-00421-6	2021	1	F14, F15, C	other		False	Table 7			False	False	*	*
24	rwe	doi:10.1007/s10290-021-00410-9	2021	2	F13, F14, F	other		False			Table 3	False	True	5,5	3,8E-08
25	rwe	doi:10.1007/s10290-021-00410-9	2021	2	F13, F14, F	other		False			Table 3	False	True	16,5	3,67E-61
26	rwe	doi:10.1007/s10290-021-00410-9	2021	2	F13, F14, F	other		False			Table 3	False	True	6,9	5,2E-12
27	rwe	doi:10.1007/s10290-021-00410-9	2021	2	F13, F14, F	other		False			Table 3	False	True	6,4	1,55E-10
28	rwe	doi:10.1007/s10290-021-00410-9	2021	2	F13, F14, F	other		False			Table 3	False	True	2	0,0455
29	rwe	doi:10.1007/s10290-021-00410-9	2021	2	F13, F14, F	other		False			Table 2	False	False	1,5	0,133614
30	rwe	doi:10.1007/s10290-021-00410-9	2021	2	F13, F14, F	other		False			Table 2	False	False	5,357143	8,45E-08
31	rwe	doi:10.1007/s10290-021-00410-9	2021	2	F13, F14, F	other		False			Table 2	False	False	3,785714	0,000153

Figure A.1: Output from artifact

Appendix B

Economic Journal P-value goldstandard

The annotated research data used in the thesis is made available through GitHub and may be accessed through the following URL:

https://github.com/LoweEri/Economic-Journal-P-value-goldstandard

Appendix C

Reflection Document - Kim

How does your study correspond to the goals of the thesis course? Why? Focus on the goals that were achieved especially well and those that were not well achieved.

While I believe all learning goals have been achieved, there are some that was done so especially well. From the beginning of the thesis we put a lot of focus on crafting a thorough background section of the field metascience and how this could relate to computer and systems science. The learning goal of analyzing and criticizing relevant scientific literature was therefore well met. I also believe that when developing our solution, we put a lot of consideration into how we could web scrape ethically, since that is an important area and a lot of damaged could possibly be done if these considerations were disregarded by us. The learning goal of reflecting about ethical aspects was therefore also well met.

How did the planning of your study work? What could you have done better?

From the start of the thesis we planned our work by during the beginning of each week defining a schedule of what was needed to be done at that point. This planning, along with having seminars with our supervisor and two other students also writing a thesis, has given a lot of structure to how we have conducted our work. Before each seminar our supervisor had planned deadlines for which parts of the thesis should be completed and handed in before the meetings. That helped in finishing the thesis on time. Since we had a clear structure of how we were going to complete out thesis from the outset, I think the planning was well done and no parts of it could be much improved. I did not rely on the milestones in SciPro very much, which possibly could have been good to do. However, we still completed the thesis on time and therefore it does not seem like an important improvement of the planning.

How does the thesis work relate to your education? Which courses

and areas have been most relevant for your thesis work?

Several courses throughout my studies at DSV has been relevant for this thesis. We got to learn a lot about design science in the course Research Methodology for Computer and Systems Science. In that course we also performed a design science project. That experience has helped a lot during the thesis work. We could quickly identify that design science would be an appropriate strategy for us to work with, and we could implement the strategy with some ease.

Furthermore, several more technical courses have been valuable for developing the artifact. Programming courses of course, but also courses in big data, data mining, and business intelligence. While not directly correlated, they have provided knowledge about working with data and development using Python as a programming language.

How valuable is the thesis for your future work and/or studies?

Choosing design science as our research strategy has been very valuable. Working with a development project has been educational, and for me it has been relevant to what I am going to work with after my studies. Before selecting this project idea for our thesis, I knew I wanted to do a thesis relating to data collection or possible business intelligence. While metascience was a new field for me, it has been very interesting to learn about. Even though I am not going to continue working in any academic field, I think performing a thesis relating to metascience has taught me more about critical thinking which I think is important in any field.

How satisfied are you with your thesis work and its results? Why?

I believe we have managed to write a thesis of good quality with results that demonstrate that this solution could be used in an actual quantitative metascience study. In that sense, it has felt meaningful to both develop the solution and to evaluate it. The evaluation results are one of the most important parts of our thesis according to me. I am very satisfied with our results and the work we have performed. It was enjoyable to work with a practical project and to use design science as a strategy.

Appendix D

Reflection Document - Lowe

How does your study correspond to the goals of the thesis course? Why? Focus on the goals that were achieved especially well and those that were not well achieved.

I believe the learning goal for the master thesis writing course has been met. We have created an academic thesis in conjunction with our supervisor, with the insights and feedback provided by other students. There is a clear connection between the work we have made and the domain of computer and system sciences in that many of the disciplines found within the science have been applied and utilized. The thesis has, in my opinion, helped to narrow the research gap within metascience by providing a solid background grounded in previous scientific research as well as providing a functioning tool that may be used within the science. Ethical considerations were analyzed, identified and then accounted for. The ethical areas identified were quite unique and, in my opinion, interesting.

How did the planning of your study work? What could you have done better?

The planning for the development of the artifact was done on a weekly basis in combination with setting goals for the writing of the thesis. A schedule was made for what milestones to meet for the project for the next week. The process was performed iteratively - meaning if a goal was not met, it was moved to the backlog and worked on later when possible.

For proofreading and external feedback, monthly seminars were held by our supervisor Hercules Dalianis and another group who were writing their bachelor's thesis. Each group presented their work and oposed each other's work. This was very beneficial, in my opinion, as it allowed for different perspectives and being made aware of outright errors and deficiencies which afterwards could be corrected for the next seminar.

As for improving the planning of the thesis work, I cannot identify any areas of improvement. The well-conducted planning, scheduling and seminars greatly

enhanced the quality of the thesis and artifact produced.

How does the thesis work relate to your education? Which courses and areas have been most relevant for your thesis work?

The thesis work and development of the artifact can be related and tied to many of the disciplines within Computer-System Sciences. For the writing, several of the academic writing courses found at DSV were beneficial for increasing the writing quality, coherence and correctness, for instance, the course FMVEK was highly beneficial.

For the methodology and research, the course MMII was relevant as the course aims to teach research methodology within the domain of computer and systems sciences and also teaches the use of design science research, which was applied in the thesis.

For the artifact development, several programming courses were highly relevant as the creation of the artefact encompassed software development. Systems modelling courses, both for creating flow charts over the software logic and operation and object-oriented modelling, were also beneficial.

How valuable is the thesis for your future work and/or studies?

The thesis, in my opinion, is very relevant for future work and studies. The artifact, as stated in the thesis, is intended to be used within meta-science research to study the phenomena of p-hacking and could be used in future academic studies. I deem the thesis as being usable in feature work scenarios as it has given me further knowledge and experience within serval disciplines.

How satisfied are you with your thesis work and its results? Why?

I think the thesis written is of high quality and structure; furthermore, the artifact we produced fulfils the requirements and functionality required for being used in future metascience research, which I think is something positive. Personally, I found it fulfilling to work with both the academic writing in conjunction with the more practical process of developing the artifact.